

Technology Stack

Project Technology Stack

Date **05 July 2026**

Team ID *********

Project Name **StudyAI – AI Powered Smart Study Companion**

Maximum Marks 3 Marks

Introduction

The StudyAI project is developed using modern web technologies and Artificial Intelligence to provide students with an intelligent study assistant. The application follows a client-server architecture where the React frontend communicates with a Flask backend through REST APIs. AI-powered content generation is performed using the Groq API and Llama language model.

Technology Stack

Layer	Technology Used	Purpose
Frontend	React.js	User Interface
Programming Language	JavaScript	Frontend Logic
Backend	Flask	REST API Development
Backend Language	Python	Server-side Processing
AI Model	Groq API (Llama Model)	AI-based Summary Generation
PDF Processing	PyMuPDF (fitz)	Extract text from PDF documents
File Storage	Local Uploads Folder	Store uploaded study materials
API Testing	Postman	Testing backend APIs
Version Control	Git	Source Code Management
Repository	GitHub	Code Hosting and Collaboration
IDE	Visual Studio Code	Development Environment

Frontend Technologies

React.js

React.js is used to build an interactive and responsive user interface. It enables component-based development, making the application modular, reusable, and easy to maintain.

HTML5

Provides the structural layout of the web application.

CSS3

Used to create an attractive, responsive, and user-friendly interface.

JavaScript (ES6)

Handles user interactions, API requests, and dynamic rendering of components.

Backend Technologies

Python

Python is the primary programming language used for implementing the backend logic.

Flask

Flask provides lightweight REST APIs for:

- File Upload
 - AI Summary Generation
 - Flashcard Generation
 - Quiz Generation
 - Study Planner
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Artificial Intelligence

Groq API

The Groq API is integrated to generate intelligent summaries from extracted study material.

Llama Language Model

The Llama large language model processes extracted text and produces concise, easy-to-understand summaries.

PDF Processing

PyMuPDF (fitz)

PyMuPDF extracts text from uploaded PDF documents with high accuracy before sending it to the AI model.

Development Tools

Visual Studio Code

Primary IDE used for developing both frontend and backend.

Git

Tracks changes made to the project source code.

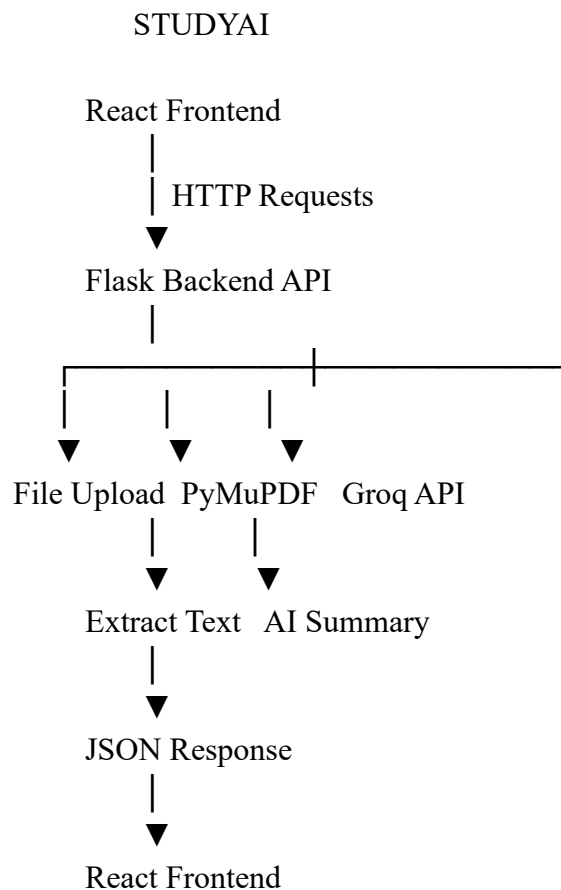
GitHub

Hosts the project repository and enables version control and collaboration.

Postman

Used for testing backend APIs before integrating them with the frontend.

System Architecture



Advantages of the Selected Technology Stack

- Lightweight and scalable architecture.
- Fast communication through REST APIs.
- Easy integration of AI services.
- Modular frontend with reusable React components.
- Efficient PDF text extraction.
- Easy deployment and maintenance.
- Open-source technologies reduce development cost.

Conclusion

The selected technology stack provides a reliable and scalable foundation for StudyAI. React ensures a responsive user interface, Flask manages backend services efficiently, PyMuPDF extracts document content accurately, and the Groq API enables high-quality AI-generated summaries. Together, these technologies create an intelligent and user-friendly study platform.